

Module 3: Lesson 4

Deep Convolutional Neural Networks: Implementation of Gramian Angular Field



Outline

- ▶ Setting up the data
- ▶ GAF for financial time series

Setting up the data

In this lesson, we implement the Gramian Angular Field (GAF) methodology that we described in Lesson 3 and illustrate its outcome for a general financial time series.

To apply GAF, we need to rearrange our time series into a 2-dimensional array where each row represents a sub-sequence of the whole time series.

Formally, we split a time series $\{x_1, x_2, \dots, x_K\}$ into sequences of T observations $\{x_t, x_{t+1}, x_{t+2}, \dots, x_{t+T}\}$, for $t \in [1, K - T]$. Then, we form the array X that stacks in rows each of these sequences:

$$X = \begin{pmatrix} x_1 & x_2 & \cdots & x_T \\ x_2 & x_3 & \cdots & x_{T+1} \\ x_3 & x_4 & \cdots & x_{T+2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{K-T} & x_{K-T+1} & \cdots & x_K \end{pmatrix}$$

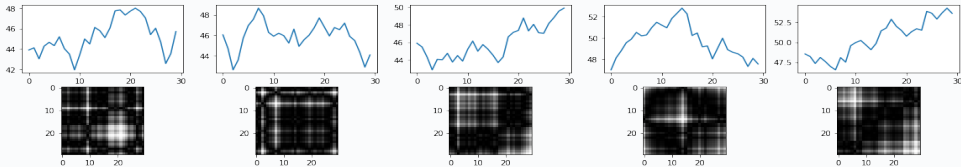
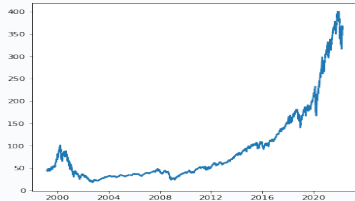
For each time series in the array above, we separately apply the GAF transformation.

GAF transformation

We implement the GAF transformation to the time series of Nasdaq prices (QQQ).

We slice the time series into sequences of 30 days and obtain the GAF image for each of them.

From the GAF representation, we train a CNN to predict the returns from Nasdaq.



Summary of Lesson 4

In Lesson 4, we have:

- ▶ Implemented GAF imaging to transform time series into images.
- ▶ Used the outputs of GAF as inputs for CNN architectures to predict stock returns.

⇒ **References for this lesson:**

Wang, Zhiguang, and Tim Oates. "Encoding Time Series as Images for Visual Inspection and Classification Using Tiled Convolutional Neural Networks." In *Workshops at the Twenty-Ninth AAAI Conference on Artificial Intelligence*. 2015.

de Vitry, Louis. "Encoding Time Series as Images." *Medium*. October 2018.

TO DO NEXT: Please go to the Jupyter Notebook associated with Lesson 4.

With this, we complete the descriptions related to convolutional neural networks. The next two modules are devoted to time series deep learning models. See you there!